

6. API Endpoints Tested

6.1 Overview of API Endpoints Testing

In this project, multiple REST API endpoints provided by **ReqRes** (<https://reqres.in>) are tested using Postman. Each endpoint represents a specific functionality such as authentication, user retrieval, creation, updating, and deletion.

The main objective of testing these endpoints is to ensure that:

- Each API performs its intended function correctly
- Proper HTTP status codes are returned
- Response data is accurate and structured properly
- APIs handle both valid and invalid requests correctly

This section explains each API endpoint in detail along with its purpose, request type, request structure, and expected response behavior.

6.2 Login API (Authentication)

Endpoint

```
POST {{base_url}}/login
```

Purpose

The Login API is used to authenticate a user and generate a token. This simulates a real-world login process in applications where users must authenticate before accessing protected resources.

Request Body

```
{
  "email": "eve.holt@reqres.in",
  "password": "cityslicka"
}
```

Expected Response

- Status Code: **200 OK**
- Response Body contains a **token**

```
{
  "token": "QpwL5tke4Pnpja7X4"
}
```

Testing Focus

- Verify successful login with valid credentials
 - Validate token is generated
 - Check response time and status code
-

6.3 Get Users API

Endpoint

GET `{{base_url}}/users?page=2`

Purpose

This API retrieves a list of users from the system. It is used to test pagination and data retrieval functionality.

Expected Response

- Status Code: **200 OK**
- Response contains:
 - Page number
 - User data list
 - Support information

Example Response Structure

```
{
  "page": 2,
  "data": [
    {
      "id": 7,
      "email": "user@example.com",
      "first_name": "Michael",
      "last_name": "Lawson"
    }
  ]
}
```

Testing Focus

- Validate list of users is returned
 - Verify correct page data
 - Ensure response structure is consistent
-

6.4 Get Single User API

Endpoint

GET {{base_url}}/users/2

Purpose

This API retrieves details of a specific user using their ID.

Expected Response

- Status Code: **200 OK**
- Returns user details

```
{
  "data": {
    "id": 2,
    "email": "janet.weaver@reqres.in",
    "first_name": "Janet",
    "last_name": "Weaver"
  }
}
```

Testing Focus

- Validate correct user data is returned
 - Ensure API works with valid user ID
 - Verify response structure
-

6.5 Create User API

Endpoint

POST {{base_url}}/users

Purpose

This API is used to create a new user in the system. It simulates adding new records in a database.

Request Body

```
{
  "name": "Tanhaji",
  "job": "QA Engineer"
}
```

Expected Response

- Status Code: **201 Created**
- Returns created user ID and timestamp

```
{
  "name": "Tanhaji",
  "job": "QA Engineer",
  "id": "123",
  "createdAt": "2026-06-18T10:00:00.000Z"
}
```

Testing Focus

- Validate user creation success
 - Check if ID is generated
 - Verify correct status code (201)
-

6.6 Update User API (PUT Request)

Endpoint

PUT `{{base_url}}/users/2`

Purpose

PUT request is used to completely update an existing user's data.

Request Body

```
{
  "name": "Tanhaji Updated",
  "job": "Senior QA"
}
```

Expected Response

- Status Code: **200 OK**
- Returns updated data

Testing Focus

- Full data update validation
 - Ensure response reflects new values
 - Validate status code
-

6.7 Update User API (PATCH Request)

Endpoint

PATCH {{base_url}}/users/2

Purpose

PATCH request is used for partial updates of user data.

Request Body

```
{
  "job": "Lead QA Engineer"
}
```

Expected Response

- Status Code: **200 OK**
- Only updated fields returned

Testing Focus

- Partial update validation
 - Ensure unchanged fields remain intact
 - Verify response correctness
-

6.8 Delete User API

Endpoint

DELETE {{base_url}}/users/2

Purpose

This API is used to delete a user from the system.

Expected Response

- Status Code: **204 No Content**
- No response body

Testing Focus

- Validate successful deletion
- Ensure correct status code (204)

- Confirm no response body is returned

6.9 Negative Test Case (User Not Found)

Endpoint

GET {{base_url}}/users/23

Purpose

This test checks how the API behaves when an invalid user ID is requested.

Expected Response

- Status Code: **404 Not Found**

Testing Focus

- Validate proper error handling
- Ensure API returns correct error code
- Check system stability for invalid inputs

6.10 Summary of API Endpoints

API Name	Method	Purpose
Login API	POST	User authentication
Get Users	GET	Fetch list of users
Get Single User	GET	Fetch specific user
Create User	POST	Add new user
Update User (PUT)	PUT	Full update
Update User (PATCH)	PATCH	Partial update
Delete User	DELETE	Remove user

6.11 Final Conclusion

The tested API endpoints cover all major CRUD operations along with authentication functionality. These endpoints help simulate real-world backend systems where data is created, retrieved, updated, and deleted.

By testing these APIs, we ensure:

- Correct functionality of each endpoint
 - Proper handling of requests and responses
 - Reliable and consistent API behavior
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